

Ang Li

Website: angli66.github.io

Email: ang6li98@gmail.com

EDUCATION

Stanford University

M.S. Computer Science, specialized in Artificial Intelligence

Sep 2023 – Apr 2025

University of California, San Diego

B.S. Computer Science, graduated with High Honors

Aug 2019 – Dec 2022

PUBLICATION

1. Chao Xu, **Ang Li**, Linghao Chen, Yulin Liu, Ruoxi Shi, Minghua Liu, Hao Su. *SpaRP: Fast 3D Object Reconstruction and Pose Estimation from Sparse Views*. Accepted to European Conference on Computer Vision (ECCV 2024). [\[arXiv\]](#)
2. Xiaoshuai Zhang, Rui Chen, **Ang Li**, Fanbo Xiang, Yuzhe Qin, Jiayuan Gu, Zhan Ling, Minghua Liu, Peiyu Zeng, Songfang Han, Zhiao Huang, Tongzhou Mu, Jing Xu, Hao Su. *Close the Optical Sensing Domain Gap by Physics-Grounded Active Stereo Sensor Simulation*. Accepted to IEEE Transactions on Robotics (T-RO 2023). [\[arXiv\]](#)

WORK EXPERIENCE

Senior Algorithm Engineer, Simulation, Sudo AI

Feb 2026 – Present

- Own the rendering engine for a large-scale robotics simulation platform, maintaining core infrastructure for embodied AI development.
- Developed a real-time Gaussian radiance-aware foreground tracing pipeline that transfers HDR Gaussian background illumination onto ray-traced foreground objects.

Robot Simulation Engineer, Hillbot

Jan 2025 – Jan 2026

- Led Vulkan rendering system development for a large-scale robotics simulation engine, with rasterization and real-time hardware ray tracing optimized for thousands of parallel environments.
- Built a production-style PBR material system with layered GGX BRDF/BTDF and multi-scattering energy compensation.
- Built a Blender-compatible node-based shading system to reuse artist-authored material graphs in the simulation engine.
- Implemented Gaussian Splatting rasterization shaders for real-time point-based neural scene rendering.

Research Intern, Hillbot

Jul 2024 – Sep 2024

- Maintained active stereo depth sensor simulation library for realistic depth and infrared signal generation.
- Optimized Vulkan rasterization pipelines for GPU efficiency and real-time rendering performance.
- Added rendering features including multiview rendering, geometry instancing, shadow catchers, reflection probes, and tone mapping.

RESEARCH EXPERIENCE

3D Object Reconstruction and Pose Estimation from Sparse Views

Stanford

Advisor: Prof. Hao Su

2024

- Developed an optimization pipeline that enhances the accuracy of coarse predicted camera poses by leveraging the generated 3D mesh.
- Significantly outperformed baseline methods in 3D reconstruction quality and pose estimation accuracy with our proposed SpaRP method for sparse-view image inputs.

Active Stereo Vision Depth Sensor Simulation

UC San Diego

Advisor: Prof. Hao Su

2022

- Designed and implemented a CUDA-based library to simulate the stereo matching module of real-world depth sensors, enhancing computational efficiency.
- Achieved superior runtime and transfer performance compared to other sim-to-real methods with our proposed depth sensor simulation pipeline.

Closed-Loop Control for Mechanical Ventilation

UC San Diego

Advisor: Prof. Ryan Kastner

2021

- Led the development of a closed-loop control circuit for a cost-effective mechanical ventilator prototype, optimizing system reliability.
- Implemented a PID controller for Pressure Control Ventilation on Arduino chips.

SELECTED PROJECTS

SAPIEN | [\[Project\]](#)

- SAPIEN is an open-source and physics-rich simulation platform for robotics and embodied AI, led by Hillbot.

SimSense | [\[GitHub\]](#)

- SimSense is a GPU-accelerated depth sensor simulator for python based on stereo matching.

TEACHING

CSE 152A: Introduction to Computer Vision

UC San Diego

Instructional Assistant

Sep 2022 – Dec 2022

- The course covers image formation, reconstruction, classification, recognition, deep learning.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Rust, GLSL, CUDA

Frameworks and Libraries: NumPy, PyTorch, OpenGL, Vulkan

Development Tools: Git, Docker, Kubernetes, Blender

Domain Knowledge: Computer Vision, Computer Graphics, Robotics, Deep Learning, Reinforcement Learning, Physics